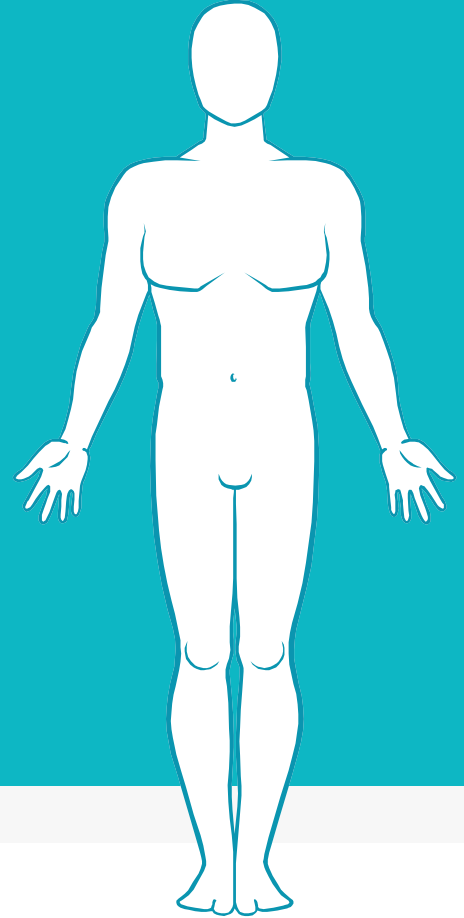


MEC E 686

Assessment and Analysis of Biomechanical Motion

Assignment 2 Guide



Body Segment Parameters

- ▶ Individual-specific body segment parameters (BSPs)
 - ▶ Mass
 - ▶ Joint centers of rotation (JCR)
 - ▶ Center of mass (COM)
 - ▶ Length
 - ▶ Moment of inertia tensor

Body Segment Parameters

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TABLE 4.1 Anthropometric Data

Segment	Definition	Segment Weight/Total	Center of Mass/ Segment Length		Radius of Gyration/ Segment Length			Density
		Body Weight	Proximal	Distal	C of G	Proximal	Distal	
Hand	Wrist axis/knuckle II middle finger	0.006 M	0.506	0.494 P	0.297	0.587	0.577 M	1.16
Forearm	Elbow axis/ulnar styloid	0.016 M	0.430	0.570 P	0.303	0.526	0.647 M	1.13
Upper arm	Glenohumeral axis/elbow axis	0.028 M	0.436	0.564 P	0.322	0.542	0.645 M	1.07
Forearm and hand	Elbow axis/ulnar styloid	0.022 M	0.682	0.318 P	0.468	0.827	0.565 P	1.14
Total arm	Glenohumeral joint/ulnar styloid	0.050 M	0.530	0.470 P	0.368	0.645	0.596 P	1.11
Foot	Lateral malleolus/head metatarsal II	0.0145 M	0.50	0.50 P	0.475	0.690	0.690 P	1.10
Leg	Femoral condyles/medial malleolus	0.0465 M	0.433	0.567 P	0.302	0.528	0.643 M	1.09
Thigh	Greater trochanter/femoral condyles	0.100 M	0.433	0.567 P	0.323	0.540	0.653 M	1.05
Foot and leg	Femoral condyles/medial malleolus	0.061 M	0.606	0.394 P	0.416	0.735	0.572 P	1.09
Total leg	Greater trochanter/medial malleolus	0.161 M	0.447	0.553 P	0.326	0.560	0.650 P	1.06
Head and neck	C7-T1 and 1st rib/ear canal	0.081 M	1.000	— PC	0.495	0.116	— PC	1.11
Shoulder mass	Sternoclavicular joint/glenohumeral axis	—	0.712	0.288	—	—	—	1.04
Thorax	C7-T1/T12-L1 and diaphragm*	0.216 PC	0.82	0.18	—	—	—	0.92
Abdomen	T12-L1/L4-L5*	0.139 LC	0.44	0.56	—	—	—	—
Pelvis	L4-L5/greater trochanter*	0.142 LC	0.105	0.895	—	—	—	—
Thorax and abdomen	C7-T1/L4-L5*	0.355 LC	0.63	0.37	—	—	—	—
Abdomen and pelvis	T12-L1/greater trochanter*	0.281 PC	0.27	0.73	—	—	—	1.01
Trunk	Greater trochanter/glenohumeral joint*	0.497 M	0.50	0.50	—	—	—	1.03
Trunk head neck	Greater trochanter/glenohumeral joint*	0.578 MC	0.66	0.34 P	0.503	0.830	0.607 M	—
Head, arms, and trunk (HAT)	Greater trochanter/glenohumeral joint*	0.678 MC	0.626	0.374 PC	0.496	0.798	0.621 PC	—
HAT	Greater trochanter/mid rib	0.678	1.142	—	0.903	1.456	—	—

*NOTE: These segments are presented relative to the length between the greater trochanter and the glenohumeral joint.

Source Codes: M, Dempster via Miller and Nelson; *Biomechanics of Sport*, Lea and Febiger, Philadelphia, 1973. P, Dempster via Plagenhoef; *Patterns of Human Motion*, Prentice-Hall, Inc. Englewood Cliffs, NJ, 1971. L, Dempster via Plagenhoef from living subjects; *Patterns of Human Motion*, Prentice-Hall, Inc., Englewood Cliffs, NJ, 1971. C, Calculated.

David A. Winter, Biomechanics and motor control of human movement, 4th Edition, Page 86



Body Segment Parameters

- ▶ **Mass of segment:** based on each individual's body mass (BM)

$$Mass_{Pelvis} = 0.142 \cdot BM$$

$$Mass_{Thigh} = 0.100 \cdot BM$$

$$Mass_{Shank} = 0.0465 \cdot BM$$

$$Mass_{Foot} = 0.0145 \cdot BM$$

- ▶ **Joint Center of Rotation (JCR):** based on the location of anatomical landmarks

Left body side: $JCR_{ground} = \frac{LMH5 + LMH1}{2}$ $JCR_{ankle} = \frac{LMM + LLM}{2}$ $JCR_{knee} = \frac{LLE + LME}{2}$

Right body side: $JCR_{ground} = \frac{RMH5 + RMH1}{2}$ $JCR_{ankle} = \frac{RMM + RLM}{2}$ $JCR_{knee} = \frac{RLE + RME}{2}$



Body Segment Parameters

- ▶ **Hip JCR:** based on pelvic landmark according to:
 - ▶ *D. Gordon et al., Research Methods in Biomechanics, 2nd Edition, Page 40*

Norm of the vector from LASIS to RASIS:

$$PW = \text{norm}({}^L_R (RASIS - LASIS))$$

Right side: $JCR.RHip = {}^G_L R \begin{bmatrix} -0.19 PW \\ -0.30 PW \\ 0.36 PW \end{bmatrix} + P_{\text{pelvic frame origin}}$

Left side: $JCR.LHip = {}^G_L R \begin{bmatrix} -0.19 PW \\ -0.30 PW \\ -0.36 PW \end{bmatrix} + P_{\text{pelvic frame origin}}$

The sequence of x, y, z in above-mentioned definition is modified according to the lab frame



Body Segment Parameters

- ▶ **COM** of each segment in the global frame:
 - ▶ *David A. Winter, Biomechanics and motor control of human movement, 4th Edition, Page 86*

In general: $\overrightarrow{COM} = \overrightarrow{JCR}_{distal} + k \times (\overrightarrow{JCR}_{proximal} - \overrightarrow{JCR}_{distal})$

or

$$\overrightarrow{COM} = \overrightarrow{JCR}_{proximal} + (1 - k) \times (\overrightarrow{JCR}_{proximal} - \overrightarrow{JCR}_{distal})$$

$$\overrightarrow{COM}_{Foot} = \overrightarrow{JCR}_{Ground} + 0.5 \times (\overrightarrow{JCR}_{Ankle} - \overrightarrow{JCR}_{Ground})$$

$$\overrightarrow{COM}_{Shank} = \overrightarrow{JCR}_{Ankle} + 0.567 \times (\overrightarrow{JCR}_{Knee} - \overrightarrow{JCR}_{Ankle})$$

$$\overrightarrow{COM}_{Thigh} = \overrightarrow{JCR}_{Knee} + 0.567 \times (\overrightarrow{JCR}_{Hip} - \overrightarrow{JCR}_{Knee})$$



Body Segment Parameters

- ▶ **Length** of each segment:

In general: $Length = norm(\overrightarrow{JCR}_{proximal} - \overrightarrow{JCR}_{distal})$

$$Length_{Foot} = norm(\overrightarrow{JCR}_{Ankle} - \overrightarrow{JCR}_{Ground})$$

$$Length_{Shank} = norm(\overrightarrow{JCR}_{Knee} - \overrightarrow{JCR}_{Ankle})$$

$$Length_{Thigh} = norm(\overrightarrow{JCR}_{Hip} - \overrightarrow{JCR}_{Knee})$$

Body Segment Parameters

- ▶ **Moment of Inertia tensor of each segment:**
 - ▶ R. Dumas et al., Adjustments to McConville et al. and Young et al. body segment inertial parameters, 2007, Journal of Biomechanics, 40 pp. 453-533

Segment	Length definition	Origin of SCS	Gender	Length <i>L</i> (in mm)	Scaling factor for mass <i>m</i> (%)	Scaling factors for position of centre of mass			Scaling factors for tensor of inertia					
						<i>X</i> (%)	<i>Y</i> (%)	<i>Z</i> (%)	<i>r_{xx}</i> (%)	<i>r_{yy}</i> (%)	<i>r_{zz}</i> (%)	<i>r_{xy}</i> (%)	<i>r_{xz}</i> (%)	<i>r_{yz}</i> (%)
Head & Neck	<i>CJC</i> to <i>HV</i>	<i>CJC</i>	F	221	6.7	-7.0	59.7	0	32	27	34	6(i)	1	1(i)
Torso	<i>CJC</i> to <i>LJC</i>	<i>CJC</i>	M	244	6.7	-6.2	55.5	0.1	31	25	33	9(i)	2(i)	3
			F	429	30.4	-1.6	-43.6	-0.6	29	27	29	22	5	5(i)
Arm	<i>SJC</i> to <i>EJC</i>	<i>SJC</i>	M	477	33.3	-3.6	-42.0	-0.2	27	25	28	18	2	4(i)
			F	243	2.2	-7.3	-45.4	-2.8	33	17	33	3	5(i)	14
Forearm	<i>EJC</i> to <i>WJC</i>	<i>EJC</i>	M	271	2.4	1.7	-45.2	-2.6	31	14	32	6	5	2
			F	247	1.3	2.1	-41.1	1.9	26	14	25	10	4	13(i)
Hand	<i>WJC</i> to midpoint between <i>MH₂</i> and <i>MH₅</i>	<i>WJC</i>	M	283	1.7	1.0	-41.7	1.4	28	11	27	3	2	8(i)
			F	71	0.5	7.7	-76.8	4.8	63	43	58	29	23	28(i)
Pelvis	<i>LJC</i> to projection of <i>HJC</i> in sagittal plane	<i>LJC</i>	M	80	0.6	8.2	-83.9	7.4	61	38	56	22	15	20(i)
			F	107	14.6	-0.9	-23.2	0.2	91	100	79	34(i)	1(i)	1(i)
Thigh	<i>HJC</i> to <i>KJC</i>	<i>HJC</i>	M	94	14.2	2.8	-28.0	-0.6	101	106	95	25(i)	12(i)	8(i)
			F	379	14.6	-7.7	-37.7	0.9	31	19	32	7	2(i)	7(i)
Leg	<i>KJC</i> to <i>AJC</i>	<i>KJC</i>	M	432	12.3	-4.1	-42.9	3.3	29	15	30	7	2(i)	7(i)
			F	388	4.5	-4.9	-40.4	3.1	28	10	28	2	1	6
Foot	<i>AJC</i> to midpoint between <i>MH₁</i> and <i>MH₅</i>	<i>AJC</i>	M	433	4.8	-4.8	-41.0	0.7	28	10	28	4(i)	2(i)	5
			F	165	1.0	27.0	-21.8	3.9	17	36	35	10(i)	6	4(i)
Alternative length and origin														
Torso	<i>C₇</i> to <i>SUP</i>	<i>SUP</i>	F	125	30.4	-41.1	-117.3	-1.9	98	93	98	76	16	19(i)
			M	139	33.3	-45.6	-112.1	-0.8	93	85	96	62	7	13(i)
Hand	<i>WJC</i> to <i>FT₃</i>	<i>WJC</i>	F	167	0.5	3.3	-32.7	2.1	27	18	25	12	10	12(i)
			M	189	0.6	3.5	-35.7	3.2	26	16	24	9	7	8(i)
Pelvis	Midpoint between <i>RASIS</i> to <i>LASIS</i>	Middle of <i>RASIS</i> and <i>LASIS</i>	F	238	14.6	-37.1	-5.0	0.1	41	45	36	15(i)	0	0

R. Dumas et al. / Journal of Biomechanics 40 (2007) 543-553



Body Segment Parameters

- Moment of Inertia tensor of each segment:

Thigh

$$I_{xx} = (0.29L)^2$$

$$I_{yy} = (0.15L)^2$$

$$I_{zz} = (0.30L)^2$$

$$I_{xy} = (0.07L)^2$$

$$I_{xz} = -(0.02L)^2$$

$$I_{yz} = -(0.07L)^2$$

Shank

$$I_{xx} = (0.28L)^2$$

$$I_{yy} = (0.10L)^2$$

$$I_{zz} = (0.28L)^2$$

$$I_{xy} = -(0.04L)^2$$

$$I_{xz} = -(0.02L)^2$$

$$I_{yz} = (0.05L)^2$$

Foot

$$I_{xx} = (0.17L)^2$$

$$I_{yy} = (0.37L)^2$$

$$I_{zz} = (0.36L)^2$$

$$I_{xy} = (0.13L)^2$$

$$I_{xz} = -(0.08L)^2$$

$$I_{yz} = 0$$

$$I = Mass. (R \times \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{xy} & I_{yy} & I_{yz} \\ I_{xz} & I_{yz} & I_{zz} \end{bmatrix} \times R^T)$$

R: the rotation matrix from Dumas' Anatomical Frame frame to our Anatomical Frame

Velocity and Acceleration

► **Angular** velocity and acceleration:

$$W = F_s \cdot ({}^G_L R_{t-1}^T \times {}^G_L R_t - I_{3 \times 3}) \quad \omega_{t-1} = \begin{bmatrix} W_{32} \\ W_{13} \\ W_{21} \end{bmatrix}$$

$$\alpha = \text{diff}(\omega) \cdot F_s \quad \alpha = \text{sgolayfilt}(\alpha)$$

$$\omega = \text{sgolayfilt}(\omega)$$

► **Linear** velocity and acceleration:

$$\vec{V} = \text{diff}(\overrightarrow{COM}) \cdot F_s \quad \vec{V} = \text{sgolayfilt}(\vec{V})$$

$$\vec{a} = \text{diff}(\vec{V}) \cdot F_s \quad \vec{a} = \text{sgolayfilt}(\vec{a})$$

diff: numerical derivative

W : spin matrix

$\vec{\omega}$: angular velocity

$\vec{\alpha}$: angular acceleration

$\vec{F}_s$: sampling rate

$\vec{V}$: linear velocity

$\vec{a}$: linear acceleration

sgolayfilt: Savitzky-Golay smoothing filtering

Velocity and Acceleration

- **Angular** velocity and acceleration:

$$W = F_s \cdot ({}^G_L R_{t-1}^T \times {}^G_L R_t - I_{3 \times 3}) \quad \omega_{t-1} = \begin{bmatrix} W_{32} \\ W_{13} \\ W_{21} \end{bmatrix} \quad \omega = \text{sgolayfilt}(\omega)$$

$$\alpha = \text{diff}(\omega) \cdot F_s \quad \alpha = \text{sgolayfilt}(\alpha)$$

- **Linear** velocity and acceleration:

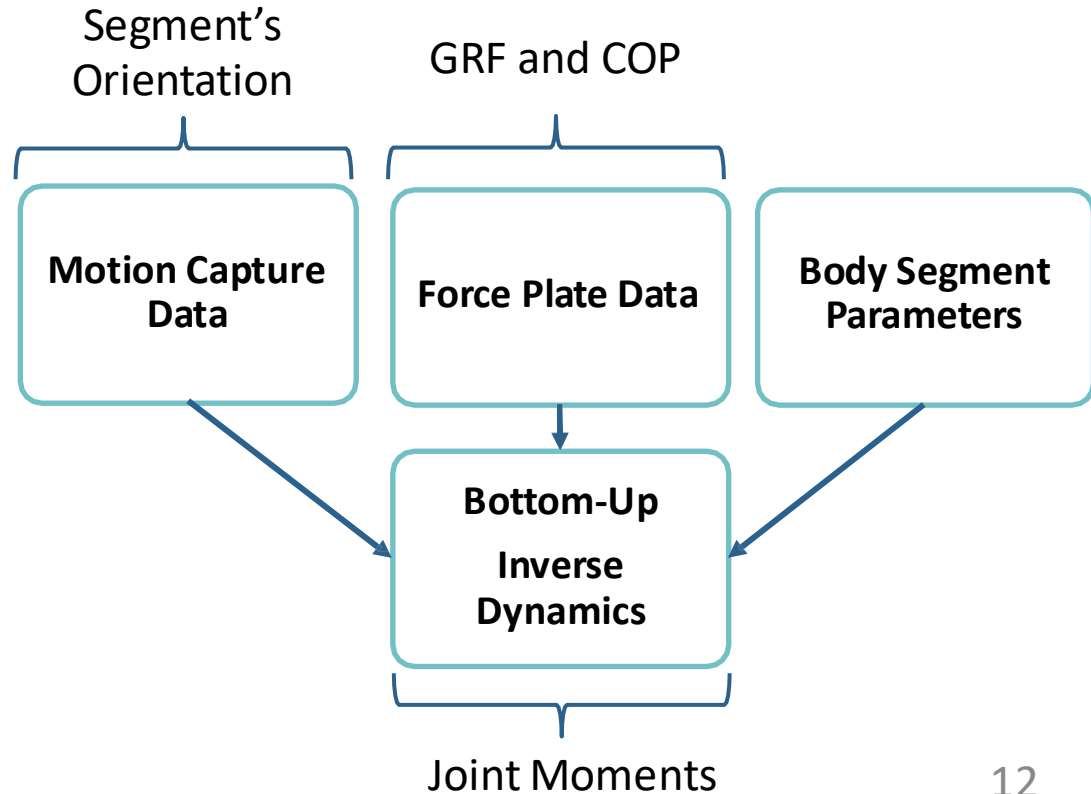
$$\vec{V} = \text{diff}(\overrightarrow{COM}) \cdot F_s \quad \vec{V} = \text{sgolayfilt}(\vec{V})$$

$$\vec{a} = \text{diff}(\vec{V}) \cdot F_s \quad \vec{a} = \text{sgolayfilt}(\vec{a})$$

Hint: The “diff” function in MATLAB has three inputs that you need to modify to address the issue you are facing. `diff(X,n,dim)` is the nth difference calculated along the dimension specified by dim.

Inverse Dynamics

Bottom-up Inverse Dynamics



COP: Center of Pressure
GRF: Ground Reaction Forces

Inverse Dynamics

$$\vec{F}_i = m_{s_{i-1}} (\vec{a}_{s_{i-1}} - \vec{g}) + \vec{F}_{i-1}$$

$$\vec{M}_{s_{i-1}} = {}^{G}_{AF}R (I_{s_{i-1}} \vec{\alpha}_{s_{i-1}} + \vec{\omega}_{s_{i-1}} \times I_{s_{i-1}} \vec{\omega}_{s_{i-1}})$$

$$\vec{M}_i = \vec{M}_{i-1} + \vec{M}_{s_{i-1}} + \left(\vec{P}_{COM_{s_{i-1}}} - \vec{P}_{JCR_i} \right) \times \vec{F}_i + \left(\vec{P}_{JCR_{i-1}} - \vec{P}_{COM_{s_{i-1}}} \right) \times \vec{F}_{i-1}$$

- **i and s_i** : joint and segment indexes
- **$\vec{M}_i$ and $\vec{F}_i$** : moment and force vectors acting on joint i
- **$\vec{a}_{s_i}$** : a segment's linear acceleration
- **$\vec{P}_{COM_{s_i}}$ and $\vec{P}_{JCR_i}$** : position of a segment's COM and JCRs
- **$I_{s_{i-1}}$, $\vec{\alpha}_{s_i}$, and $\vec{\omega}_{s_i}$** : moments of inertia, angular acceleration and velocity of the segment expressed in the segment-fixed frame
- **${}^{G}_{AF}R$** is the rotation matrix from the segment-fixed frame to the lab-fixed frame

Inverse Dynamics

$i = 1:$

$$\vec{F}_{ankle} = m_{foot} (\vec{a}_{foot} - \vec{g}) + \vec{F}_{grf}$$

$$\vec{M}_{foot} = {}_{foot}^G R (I_{foot} \vec{\alpha}_{foot} + \vec{\omega}_{foot} \times I_{foot} \vec{\omega}_{foot})$$

$$\vec{M}_{ankle} = \vec{M}_{forceplate} + \vec{M}_{foot} + (\vec{P}_{COM_{foot}} - \vec{P}_{JCR_{ankle}}) \times \vec{F}_{ankle} + (\vec{P}_{cop} - \vec{P}_{COM_{foot}}) \times \vec{F}_{grf}$$

$i = 2:$

$$\vec{F}_{knee} = m_{shank} (\vec{a}_{shank} - \vec{g}) + \vec{F}_{ankle}$$

$$\vec{M}_{shank} = {}_{shank}^G R (I_{shank} \vec{\alpha}_{shank} + \vec{\omega}_{shank} \times I_{shank} \vec{\omega}_{shank})$$

$$\vec{M}_{knee} = \vec{M}_{ankle} + \vec{M}_{shank} + (\vec{P}_{COM_{shank}} - \vec{P}_{JCR_{knee}}) \times \vec{F}_{knee} + (\vec{P}_{JCR_{ankle}} - \vec{P}_{COM_{shank}}) \times \vec{F}_{ankle}$$

Hint: The signs before the terms F_x , F_y , and F_z depends on whether they represent the GRF or the forces that are applied by the participant to the force plate(-GRF). You need to consider this while writing your code and make sure that you are using the GRF in your calculations.

Inverse Dynamics

$i = 3$:

$$\vec{F}_{hip} = m_{thigh} (\vec{a}_{thigh} - \vec{g}) + \vec{F}_{knee}$$

$$\vec{M}_{thigh} = {}_{thigh}^G R (I_{thigh} \vec{\alpha}_{thigh} + \vec{\omega}_{thigh} \times I_{thigh} \vec{\omega}_{thigh})$$

$$\vec{M}_{hip} = \vec{M}_{knee} + \vec{M}_{thigh} + \left(\vec{P}_{COM_{thigh}} - \vec{P}_{JCR_{hip}} \right) \times \vec{F}_{hip} + \left(\vec{P}_{JCR_{knee}} - \vec{P}_{COM_{thigh}} \right) \times \vec{F}_{knee}$$

Inverse Dynamics

Joint powers can be calculated as follows. Note that all joint moment and angular velocities in each equation must be expressed in the same coordinate system, e.g., the global coordinate system, or the proximal segment's local coordinate system.

$$P_{ankle} = \vec{M}_{ankle} \cdot \vec{\omega}_{shank} - \vec{M}_{ankle} \cdot \vec{\omega}_{foot}$$

$$P_{knee} = \vec{M}_{knee} \cdot \vec{\omega}_{thigh} - \vec{M}_{knee} \cdot \vec{\omega}_{shank}$$

$$P_{hip} = \vec{M}_{hip} \cdot \vec{\omega}_{pelvis} - \vec{M}_{hip} \cdot \vec{\omega}_{thigh}$$